Sophie-Marie Wind[1], Julian Varghese[1], Carolin Walter[1]

[1]Institute of Medical Informatics, University of Münster, Münster, 48149, Germany

Institut für Medizinische Informatik

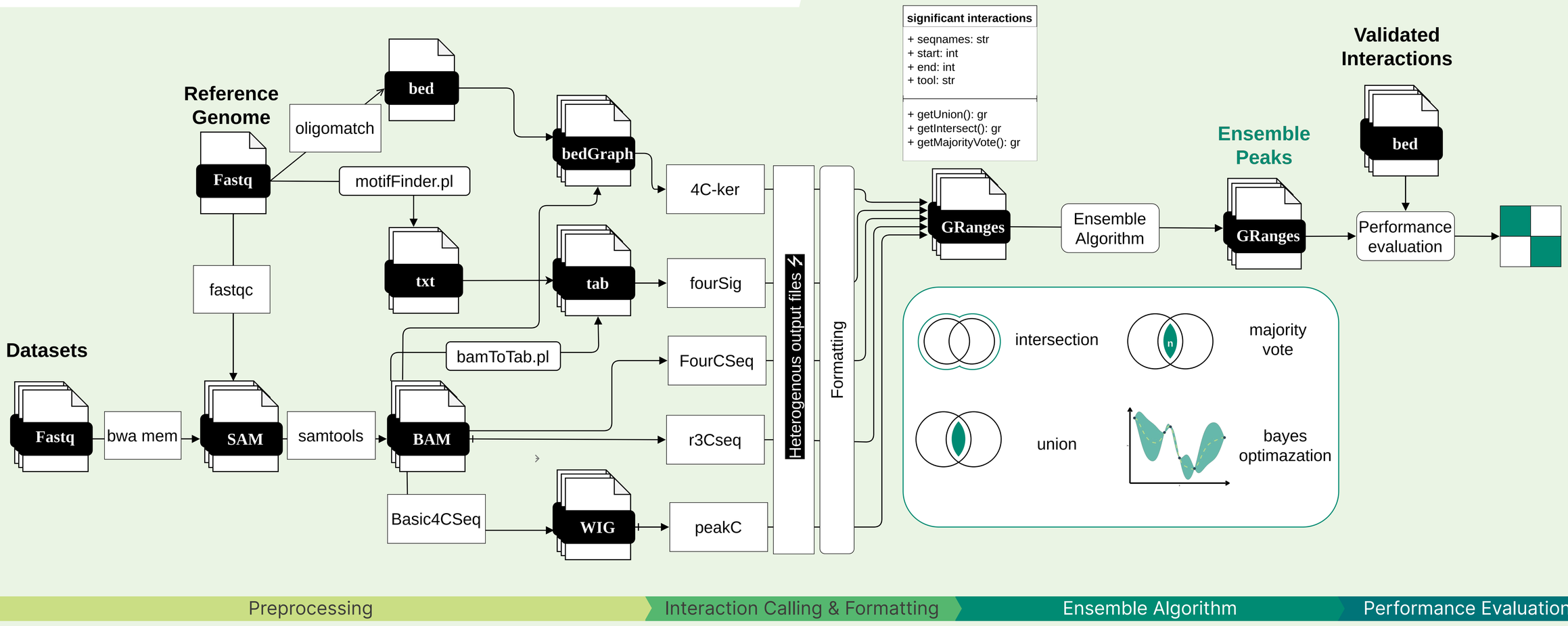
Benchmarking of 4C-seq algorithm combinations on published and simulated data

INTRODUCTION

Chromatin conformation alterations can be associated with cancer. As they are reversible, they may be potential targets for tumor therapy [1]. Circular chromosome conformation capture sequencing (4C-seq) can provide insights into the 3D structure of selected genomic loci. However, the analysis of 4C-seq data is highly complex and prone to biases. At present, several algorithms are available, but none of them is suitable for all use cases [2]. To uncover potential synergies between the algorithms, we present a comprehensive benchmarking study examining different combination strategies of existing 4C-seq algorithms.

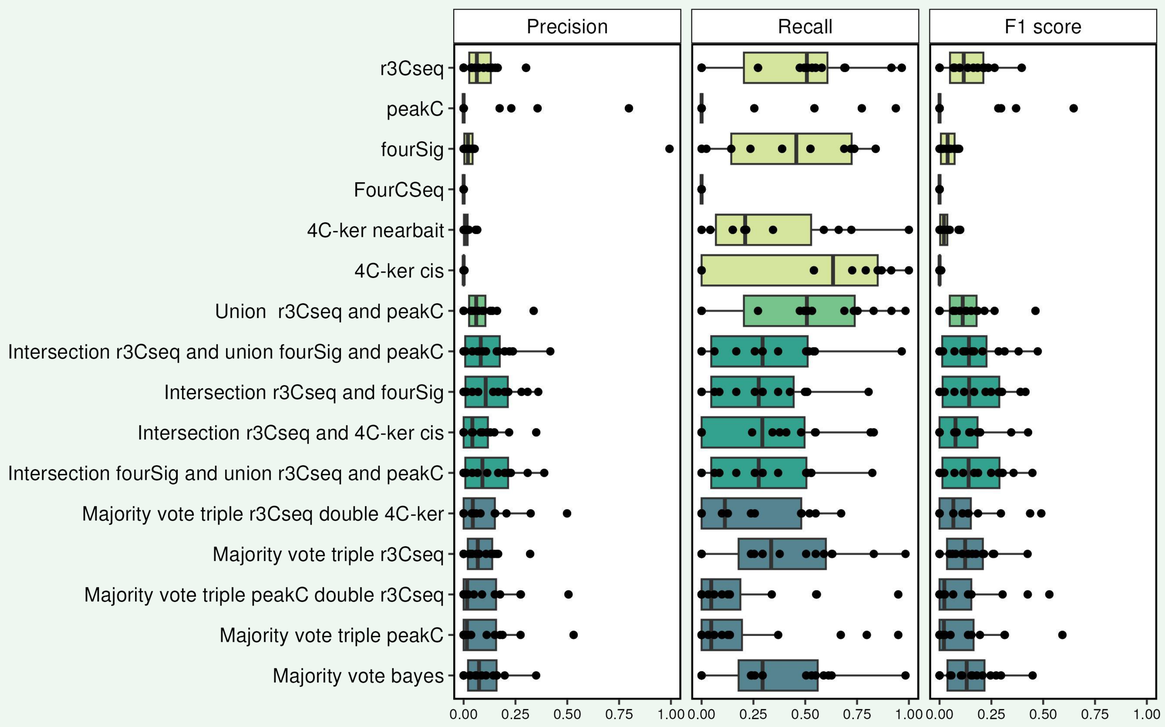
METHODS

Public and Basic4CSim [2] simulated 4C-seq datasets were preprocessed and analyzed using: *peakc* [3], *r3Cseq* [4], *fourSig* [5], F*ourCSeq* [6] and *4C-ker* [7]. The results were combined using different operations: union, intersection, majority voting and weighted approaches (Bayesian optimization used to determine optimal weights). The performances were evaluated using precision, recall and F1 score.



RESULTS

Across all analyzed datasets, *r3Cseq* performed best in terms of median precision and F1 score. *cis* interactions of *4C-ker* showed the highest median recall. Even the simple intersect of *r3Cseq* and *fourSig* outperformed all individual tools regarding the median F1 score. Simulated data indicated similar trends and results.



Benchmarking results for near-cis interactions in published datasets. Boxplots illustrating precision, recall and F1 score metrics for individual 4C-seq analysis tools and various combinations of these [3, 4, 5, 6, 7].

CONCLUSION

The results underscore the tremendous potential of ensemble approaches to identify statistically significant chromatin interactions at selected genomic loci. The development of more sophisticated ensemble approaches holds substantial promise for better leveraging the synergies among existing algorithms.

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UKM Universitätsklinikum Münster

Universität Münster

Contact: sophie.wind@uni-muenster.de

medizinische fakultät Westfälische Wilhelms-Universität Münster